

ECO403

Assignment 1 (Fall 2024)

Solution By Vu360

**Solution:****a) Cost of Basket of Goods**

To calculate the cost of the basket, multiply quantities by prices for each year.

Year	Pencils (20)	Books (16)	Burgers (30)	Total Cost
2021	$20 \times 50 = 1000$	$16 \times 100 = 1600$	$30 \times 150 = 4500$	7100
2022	$20 \times 55 = 1100$	$16 \times 110 = 1760$	$30 \times 170 = 5100$	7960
2023	$20 \times 60 = 1200$	$16 \times 140 = 2240$	$30 \times 190 = 5700$	9140

b) Consumer Price Index (CPI)

Using 2021 as the base year (CPI = 100), calculate CPI for subsequent years.

Year	CPI Calculation	CPI
2021	$(7100 / 7100) \times 100 = 100$	100
2022	$(7960 / 7100) \times 100 = 112.11$	112.11
2023	$(9140 / 7100) \times 100 = 128.73$	128.73

c) Inflation Rate

Calculate inflation rates for 2022 and 2023 relative to the previous year.

Group 1: <https://chat.whatsapp.com/F5zhJsJtf5yL4RjDODz1fL>
 Group 2: <https://chat.whatsapp.com/HJYBCFqTTPx3QYAHhOLSa2>

Year	Inflation Rate Calculation	Inflation Rate
2022	$((112.11 - 100) / 100) \times 100 = 12.11\%$	12.11%
2023	$((128.73 - 112.11) / 112.11) \times 100 = 14.79\%$	14.79%